

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a5d9a8392540184cc.jsonl`  
- SHA-256: `d2b8bb557beabb2d0c7483d5fb0151522de802a7adc14a46982dc29bf3bd5ec0`  
- Source modified: `2026-05-31T04:26:24+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:25:31.898Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"On table extraction accuracy (table edit-distance, lower is better), the PP-StructureV3 OCR+table-structure pipeline outperforms large general-purpose VLMs and pipeline competitors: it scores 0.159 (EN) / 0.109 (ZH) on Table Edit, versus Qwen2.5-VL-72B at 0.341/0.262, GOT-OCR at 0.459/0.52, Marker-1.2.3 at 0.619/0.685, MinerU-0.9.3 at 0.18/0.344, and Mathpix at 0.243/0.32. This supports a deterministic-pipeline-over-VLM recommendation for dense tables."

****Source:**** <https://github.com/PaddlePaddle/PaddleOCR/blob/main/docs/version3.x/algorithm/PP-StructureV3/PP-StructureV3.en.md> (primary)

****Supporting quote:**** "PP-StructureV3 scores: ... Table Edit: EN 0.159, ZH 0.109 ...

Qwen2.5-VL-72B: ... Table 0.341 (EN), 0.262 (ZH) ... GOT-OCR: ... Table 0.459 (EN), 0.52 (ZH) ... Marker-1.2.3: ... Table 0.619 (EN), 0.685 (ZH)"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:25:34.832Z)

I'll verify this claim by examining the source and checking for contradicting evidence. Let me start by loading the necessary tools and fetching the primary source.

3. assistant (2026-05-31T04:25:38.021Z)

I'll fetch the primary source to verify the exact benchmark numbers claimed.

4. user (2026-05-31T04:25:43.516Z)

TableEdit Scores from OmniDocBench

Benchmark: OmniDocBench (from "OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations")

Metric: TableEdit (lower scores indicate better performance)

Method	EN	ZH
PP-StructureV3	0.159	0.109
Qwen2.5-VL-72B	0.341	0.262
GOT-OCR	0.459	0.52
Marker-1.2.3	0.619	0.685
MinerU-0.9.3	0.18	0.344
Mathpix	0.243	0.32

PP-StructureV3 achieves the lowest (best) TableEdit scores among all methods tested, particularly excelling on Chinese documents (0.109).

5. assistant (2026-05-31T04:25:47.055Z)

The primary source confirms the exact numbers in the claim. All six data points match verbatim. Now let me verify the source quality and check for contradicting evidence. The key vulnerability in this claim is the interpretation: does this benchmark actually "support a deterministic-pipeline-over-VLM recommendation for dense tables"? Let me investigate.

6. user (2026-05-31T04:25:55.125Z)

Web search results for query: "OmniDocBench table recognition benchmark composition English Chinese document types financial"

Links: [{"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/opendatalab/OmniDocBench?tab=readme-ov-file"}, {"title": "GitHub - MASLINQUE/omnidocbench: A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/MASLINQUE/omnidocbench"}, {"title": "GitHub - marklabz/omnidocbench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/marklabz/omnidocbench"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/opendatalab/OmniDocBench"}, {"title": "OmniDocBench: Benchmarking Diverse PDF Document Parsing with", "url": "https://openaccess.thecvf.com/content/CVPR2025/papers/Ouyang_OmniDocBench_Benchmarking_Diverse_PDF_Document_Parsing_with_Comprehensive_Annotations_CVPR_2025_paper.pdf"}, {"title": "OmniDocBench v1.5 Benchmark", "url": "https://www.emergentmind.com/topics/omnidocbench-v1-5"}, {"title": "OmniDocBench Leaderboard: PDF Document Parsing SOTA | CodeSOTA | CodeSOTA", "url": "https://www.codesota.com/browse/computer-vision/document-parsing/omnidocbench"}, {"title": "1 Performance of GLM-OCR on OmniDocBench v1.5.", "url": "https://arxiv.org/html/2603.10910v1"}]

Based on the search results, here's information about OmniDocBench's table recognition benchmark composition:

Benchmark Composition

OmniDocBench includes 1651 PDF pages, covering 10 document types, 5 layout types, and 5 language types. It encompasses a wide range of content, including academic papers, financial reports, newspapers, textbooks, and handwritten notes.

Language Coverage

Currently, only Chinese and English text are included in text evaluation.

Table Recognition Annotations

OmniDocBench contains bounding box information for tables on each PDF page along with corresponding table recognition annotations, making it suitable as a benchmark for table recognition evaluation. The table annotations are available in both HTML and LaTeX formats, with this repository currently providing examples for HTML format evaluation.

Rich Annotation Details

OmniDocBench features rich annotations, containing 28 block-level annotations (text paragraphs, headings, tables, etc.) and 4 span-level annotations (text lines, inline formulas, subscripts, etc.). All text-related annotation boxes include text recognition annotations, formulas contain LaTeX annotations, and tables include both LaTeX and HTML annotations.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-05-31T04:25:57.389Z)

Web search results for query: "PP-StructureV3 OmniDocBench table edit distance benchmark comparison Qwen2.5-VL"

Links: [{"title": "GitHub - marklabz/omnidocbench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/marklabz/omnidocbench"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/opendatalab/OmniDocBench?tab=readme-ov-file"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/opendatalab/OmniDocBench"}, {"title": "DocPTBench: Parsing & Translation Benchmark", "url": "https://www.emergentmind.com/topics/docptbench"}, {"title": "Real5-OmniDocBench: A Full-Scale Physical Reconstruction Benchmark for Robust Document Parsing in the Wild", "url": "https://arxiv.org/html/2603.04205v1"}, {"title": "OmniDocBench v1.5 Benchmark", "url": "https://www.emergentmind.com/topics/omnidocbench-v1-5"}, {"title": "OmniDocBench v1.0 Benchmark", "url": "https://www.emergentmind.com/topics/omnidocbench-v1-0"}, {"title": "OmniDocBench Leaderboard: PDF Document Parsing SOTA | CodeSOTA | CodeSOTA", "url": "https://www.codesota.com/browse/computer-vision/document-parsing/omnidocbench"}, {"title": "PP-StructureV3 Introduction - PaddleOCR Documentation", "url": "https://paddlepaddle.github.io/PaddleOCR/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html"}]

Based on the search results, I found relevant information about PP-StructureV3, OmniDocBench, and Qwen2.5-VL benchmarking:

As of July 31, 2025, PP-StructureV3 and Qwen2.5-VL-7B were added for model evaluation in OmniDocBench. The benchmark updated the Edit Distance calculation for the Table section to use normalized ground truths and predictions.

PP-StructureV3 is listed among the officially aligned baseline models verified in the original OmniDocBench report, alongside Qwen3-VL-235B, Qwen2.5-VL-72B, Gemini-2.5 Pro, and MinerU-1.8.2.

The OmniDocBench benchmark itself includes 1651 PDF pages, covering 10 document types, 5 layout types, and 5 language types, and uses Tree Edit Distance Score for evaluating table structure, where higher is better.

For direct comparative benchmark numbers between PP-StructureV3 and Qwen2.5-VL on the table edit distance metric specifically, the search results don't contain the detailed numerical comparison table, but both models have been evaluated within the OmniDocBench framework.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-05-31T04:26:02.654Z)

Several important findings emerge. The numbers are verbatim-accurate, and OmniDocBench is a legitimate CVPR 2025 peer-reviewed benchmark. But there are some nuances I need to dig into:

- 1. The benchmark is run by PaddleOCR (the maker of PP-StructureV3) on their own documentation page ? potential self-reporting bias.
- 2. There's a critical methodological note: "The benchmark updated the Edit Distance calculation for the Table section to use normalized ground truths and predictions" as of July 31, 2025. This raises the question of whether the numbers in the PaddleOCR doc use the same methodology as the official leaderboard.
- 3. OmniDocBench is general document parsing (academic papers, financial reports, newspapers, textbooks, handwritten notes) ? not specifically dense bank-statement tables.

Let me cross-check the official OmniDocBench leaderboard to see if these numbers match independent reporting, and whether newer VLMs have changed the picture.

9. user (2026-05-31T04:26:12.272Z)

Table Recognition Scores from OmniDocBench Leaderboard

Based on the webpage content, I found table recognition evaluation results, but **not all requested models have table-specific scores displayed**.

Available Table TEDS Scores (Higher is Better)

The metric used is **TEDS (Tree Edit Distance based Similarity)**, where **higher scores indicate better performance**.

- From the end-to-end evaluation leaderboard showing Overall scores:
- **MinerU2.5-Pro**: 93.42 (Table TEDS)
 - **PaddleOCR-VL-1.5**: 91.67
 - **GOT-OCR**: Listed in formula/text sections but specific table score not shown in end-to-end table
 - **Qwen2.5-VL-72B**: Not in the main end-to-end leaderboard
 - **Qwen3-VL-235B**: 83.07
 - **Marker**: 65.77
 - **Gemini 3 Pro**: 89.15

Component-Level Table Results

- A separate component-level evaluation shows:
- **RapidTable**: 82.5 overall (OCR-based, highest)
 - **PaddleOCR**: 73.6 overall (pipeline tool)
 - **StructEqTable**: 75.8 (expert VLM)

Pattern Observed

The data suggests **specialized/pipeline methods like RapidTable and PaddleOCR outperform general VLMs** on table extraction tasks, supporting your hypothesis about OCR-pipeline superiority for tables.

10. user (2026-05-31T04:26:12.481Z)

OmniDocBench Leaderboard: Table Recognition Performance

Top Overall Rankings

The leaderboard shows **GLM-OCR** leading with a 94.62 composite score, followed by **PaddleOCR-VL** at 92.86 and **PaddleOCR-VL 0.9B** at 92.56.

Table-Specific Performance

For table recognition specifically, the content identifies these leaders:

- **PaddleOCR-VL**: 93.52 Table TEDS score
- **Qianfan-OCR**: 91.02 Table TEDS score
- **Mistral OCR 3**: 70.88 Table TEDS score

Pipeline vs. VLM Comparison

The document explicitly notes that "pipeline methods like PaddleOCR-VL and MinerU still hold the top spots, suggesting that specialized architectures remain valuable for structured document understanding." This confirms pipeline approaches outperform general vision-language models on table tasks.

2025-2026 Models

The leaderboard does not list PP-StructureV3 in the evaluated models. The newest entries mentioned are from 2026, including **PaddleOCR-VL** variants, but the document doesn't provide comparative data showing whether newer 2025-2026 models specifically beat PP-StructureV3 on table extraction.

11. assistant (2026-05-31T04:26:18.983Z)

This is illuminating. The independent OmniDocBench leaderboard data corroborates the core directional finding: pipeline/OCR-specialist methods (PaddleOCR-VL, MinerU, RapidTable) generally hold the top spots on table TEDS, beating general-purpose VLMs. One source explicitly states "pipeline methods like PaddleOCR-VL and MinerU still hold the top spots, suggesting that specialized architectures remain valuable for structured document understanding."

However, I need to assess a few key concerns for the claim's strength. Note that the leaderboard now uses TEDS (higher better) while the PaddleOCR doc uses "Table Edit" (edit distance, lower better) ? these are different but related metrics. The claim is specifically about the "Table Edit" numbers, which match the source exactly. Let me verify whether the specific PaddleOCR-reported numbers have been disputed or whether the methodology change affected them, and check the actual PaddleOCR doc again for context on test conditions.

12. user (2026-05-31T04:26:22.825Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: <https://paddlepaddle.github.io/PaddleOCR/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html>

Redirect URL: <http://www.paddleocr.ai/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html>

Status: 301 Moved Permanently

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:

- url: "http://www.paddleocr.ai/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html"
- prompt: "Find the benchmark comparison table. (1) What exact Table Edit scores are reported for PP-StructureV3, Qwen2.5-VL-72B, GOT-OCR, Marker-1.2.3, MinerU-0.9.3, Mathpix (EN and ZH)? (2) Is there any disclaimer about who ran the benchmark, the test conditions, or whether results were self-measured by PaddlePaddle? (3) What is the source/citation given for the comparison numbers ? does it cite the OmniDocBench paper/leaderboard or are these PaddlePaddle's own measurements?"

13. user (2026-05-31T04:26:23.971Z)

Web search results for query: "PaddleOCR PP-StructureV3 benchmark self-reported overstated VLM

table extraction criticism"

You've hit your session limit · resets 1:30pm (Asia/Calcutta)

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. assistant (2026-05-31T04:26:24.668Z)

You've hit your session limit · resets 1:30pm (Asia/Calcutta)